AN ECLECTIC TECHNOLOGIES FOUNDATIONAL DEFINITION

FD-EV — The Envelope

Foundational Definition: The Pre-Authorized Envelope, Its Reservation, and the Renaming of the Operational Sense

Version 0.2 — DRAFT · 2 October 2026

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Designation: FD-EV
Family: Foundational Definitions
Version: 0.2 — DRAFT (issued in the October 2026 Edition; supersedes v0.1)
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Audience: Standards committee reviewers; regulatory agency technical staff; engineering staff at regulated utilities and industrial facilities; suppliers of ICS/OT products incorporating machine learning or generative AI
Parent: None. This instrument defines. See §2.
Companion root definitions: FD-BL (the Bright Line); FD-LD (layer decomposition and determinism); FD-BR (the Broker); FD-LS (local sovereignty)
Determinations that rely on this term: FD-BL-D1 (mode of discharge) and FD-BL-D2 (indication integrity), which use "pre-authorized envelope" in the reserved sense

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1. Purpose and the Defect This Instrument Corrects

When this instrument first issued, one word, "envelope," was carrying two distinct jobs in the corpus, and a reader could not always tell which. In one job it names a region within which an above-line action is **authorized in advance** — the object a qualified human defines, whose completeness proof is

accepted, and a change to which is categorically controlled. In the other it names a region of normal operation whose **violation is a runtime event to detect and report** — the object an anomaly detector watches, whose approach triggers human review.

These are different kinds of thing. One is an object of authorization; the other is an object of monitoring. When the same word names both, a supplier reading a requirement about "the envelope" cannot tell whether the requirement is a governance obligation (who may change it, and how) or an operational one (what to detect, and when to alarm), and an instrument that mixes the two in adjacent clauses — as several do — propagates the ambiguity at every revision.

FD-BL v0.2 §6.3 recorded this collision and named the remedy: the energy sector already reserved "Bright Line" for its governance sense and renamed the colliding uses, and "the same reservation is required here." This instrument is that reservation. It reserves "envelope" for the authorization sense, renames the operational sense, and states the test that tells them apart.

2. Authority, Family, and Convention

2.1 Precedence

This instrument owns the definition of "envelope" for all documents in the corpus.

Foundational Definitions take precedence over sector frameworks, facility class documents, application guides, vendor companions, annexes, and implementation guidance. A sector framework uses the term *within* its sector; it does not define the term. This is the tier rule stated at FD-BL §2.1 and it applies here without exception.

Every other use of "envelope" is a use of the reserved term or the renamed one. A restatement may be expressed in sector vocabulary and may apply the term to sector-specific regions. It shall not use "envelope" for the operational sense once this instrument issues. Where an existing instrument does, this instrument governs and the varying text shall be conformed at that instrument's next revision. Propagation is by reference to this instrument, not by restating its text.

2.2 The Foundational Definitions family

FD-EV is a member of the Foundational Definitions family. A concept belongs in this family when the correctness of other documents depends on agreeing what kind of thing it is (FD-BL §2.2). "Envelope" meets that test directly: FD-BL-D1 and FD-BL-D2 are correct only if a reader knows the envelope they cite is the object of authorization — the region whose completeness proof is accepted under R2 and whose change is excluded from advance discharge under R3 — and not a monitored operating limit. Because the determinations' correctness turns on which kind of thing "envelope" names, the term is definitional and belongs to this family.

FD-EV is a companion root to FD-BL, FD-LD, FD-BR and FD-LS. FD-BL owns placement; FD-LD owns layer decomposition and determinism; FD-EV owns the envelope. None owns the others' vocabulary.

2.3 Declaration convention

Consistent with FD-BL §2.3, each instrument that uses either term shall make the reservation visible: it shall state, once, that "envelope" is reserved for the pre-authorized (authorization) sense and that the operational sense is named the operating band. This mirrors the one-line "Bright Line is reserved exclusively for the ML/AI governance boundary" notice already carried in the oil and gas series. The

declaration exists because the absence of it is the cause of the collision at §1: a document written in operational vocabulary will reach for "envelope" in its engineering sense without noticing it has crossed into the reserved one.

2.4 What this instrument does not alter

This instrument is definitional. It changes terminology, not substance. It does not alter any requirement of FD-BL-D1 or FD-BL-D2 — R1–R3, the two-object model, the completeness proof, the exclusion of envelope changes from advance discharge, and the indication-integrity conditions I1–I3 are unchanged in force; only the name of the operational sense changes where it appears. It states no new operational requirement for the systems it describes. Renaming the operational sense does not downgrade any monitoring, setback, or encroachment-detection obligation; those obligations continue unchanged under their new name.

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3. The Two Senses and Why They Collide

Sense G — the pre-authorized envelope (an object of authorization). A region within which an above-line action is authorized in advance. A qualified human, the Command Broker discharging in advance (FD-BR §4.2), defines it, including any setback from protective setpoints; its acceptance set is proven complete and sound under FD-BL-D1 R2; and a change *to* it is categorically excluded from advance discharge under R3, routing instead to in-the-moment authorization or offline requalification. Here the region is governed: the questions it answers are *who authorized it*, *on what proof*, *and who may change it*.

Sense O — the operating band (an object of monitoring). A region of normal operation whose boundary is a runtime limit, and whose violation or approach is an event to detect and report. An anomaly detector watches for systematic encroachment toward the boundary; a detected excursion requires human review before operation continues. Here the region is monitored: the questions it answers are *where are the limits*, *and what do we do when operation drifts toward them*.

Why they collide. DBA-GAS-FC2 uses both senses in adjacent clauses about the same physical region: *"APC logic may not autonomously adjust the operating envelope toward an SIS trip point without a human-authorized MOC process"* is Sense G — the region as an authorization object, whose change is an R3 event — while *"anomaly detection for systematic envelope encroachment … requires human review"* is Sense O — the region as a monitored limit whose approach is detected. (APC is advanced process control, SIS a safety instrumented system, and MOC management of change.) The same word does authorization work in one clause and monitoring work in the next. Nothing in the word marks the switch, so the reader must infer it. This instrument removes the inference.

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4. The Definition

4.1 What kind of thing an envelope is

An **envelope** is a pre-authorized region of action.

It is an object of authorization, not of monitoring. What makes a region an envelope is that an above-line action inside it has been authorized in advance by a qualified human on an accepted proof, and that

a change to the region is itself a governed act in the sense FD-LS §3.2 defines: a deliberate, authenticated act by a qualified human, recorded, which no automated element can perform or complete alone. An envelope is the thing FD-BL-D1 permits advance discharge *within*, and the thing FD-BL-D1 R3 forbids advance discharge *of a change to*.

A region that has none of this apparatus — no advance authorization, no accepted completeness proof, no controlled change — is not an envelope in the reserved sense, however it has been labeled. It is an operating band (§4.3).

4.2 The reservation

The term **"envelope," and the compound "pre-authorized envelope," are reserved for Sense G** — the object of authorization defined at §4.1 — throughout the corpus. FD-BL-D1 and FD-BL-D2 already use the term in exactly this sense; their usage is conforming and unchanged.

Bare "envelope," where context does not already fix the sense, shall be read as the reserved sense. An instrument that needs the operational sense shall not reach for "envelope"; it shall use the renamed term at §4.3.

4.3 The rename

The operational sense — Sense O, the region of normal operation whose violation is a runtime event to detect and report — is renamed the **operating band**. Its boundary is an **operating limit**; the approach of operation toward that limit is **encroachment**; a crossing is an **operating-limit excursion**. "Operating envelope," used in the operational sense, is renamed "operating band" wherever it appears; "envelope encroachment," in that sense, is renamed "operating-band encroachment"; and monitoring, setback, and re-validation obligations attach to the operating band under these names unchanged.

Where a single region is genuinely both — a pre-authorized envelope whose operating limits are also monitored at runtime, as in the gas-processing case — the two roles are named separately: it is a **pre-authorized envelope** when the sentence concerns its authorization or its change (Sense G), and its **operating limits / operating band** when the sentence concerns detecting encroachment toward its boundary (Sense O). The physical region is one; the two names mark which question is being asked of it.

A sector term may stand in for "operating band." Where a sector already carries an established term for the designed operating region, such as "operating limits" or a licensing term of art, that term may be adopted for that sector in place of "operating band," provided it is stated once, used consistently, and is not "envelope."

4.4 The test that tells them apart

For any use of "envelope," ask what the sentence requires of the region:

- If it concerns **who authorized the region, the proof accepted for it, or who may change it** — it is a **pre-authorized envelope** (keep "envelope"). Changing it is governed by FD-BL-D1 R3; its completeness is governed by R2.
- If it concerns **where the operating limits are, or what to detect and report when operation approaches or crosses them** — it is the **operating band** (rename). Encroachment toward its limits is monitored; an excursion triggers review.

The test is the kind of obligation, not the physical region. A region that carries both kinds of obligation carries both names, each used where its obligation is in view (§4.3).

5. Worked Examples

Illustrative; sector variants may be substituted without altering the definition.

A gas-processing APC region is defined with a human-authorized setback from the SIS trip points, and autonomous movement of that region toward a trip point requires an MOC. As the object of that authorization and change control, it is the **pre-authorized envelope** (Sense G, keeps the term); FD-BL-D1 R3 governs changes to it. The same region's boundary is watched for systematic encroachment, and a detected excursion requires human review before further adjustment. In that monitoring role it is the **operating band** with its **operating limits** (Sense O, renamed). One region, two names, each where its obligation is in view.

A control-system document states that the plant is designed to run "within its operating envelope," meaning the designed range of normal conditions, with alarms on excursion. There is no advance authorization apparatus here — no accepted completeness proof, no R3 change control. This is the operational sense throughout: it is the **operating band**, and "operating envelope" is renamed.

FD-BL-D1 permits advance discharge of the FD-BL §4.5 authorization for an above-line action inside a valid **pre-authorized envelope** under R1–R3. This is the reserved sense, unchanged: the envelope is the object the qualified human authorized on an accepted proof, and a change to it is excluded from advance discharge. No rename applies.

6. Relationship to the Determinations and the §6 Set

- **FD-BL §6.3** was resolved by this instrument. Since FD-BL v0.3, §6.3 has been a pointer here, and this instrument carries the reservation and its sweep. With §6.1 (FD-BL-D1) and §6.2 (FD-BL-D2) already resolved, FD-EV closes the third and last of the FD-BL §6 open items.
- **FD-BL-D1 and FD-BL-D2** use "pre-authorized envelope" in the reserved sense; their usage is conforming. The "held pending FD-EV" envelope annotations each carries are discharged: they cite FD-EV for the reserved term and drop the reservation. No substantive clause of either determination changes.
- **Encroachment / anti-ratcheting** language (FD-BL-D1 R2; DBA-GAS-FC2; the FD-BL-D2 I3 corroboration path) concerns monitoring the approach to a boundary; it does not use "envelope" in the operational sense, and takes the operating-band vocabulary where a monitored region is named. Its role under R2 (bounding selection over trajectory and rate) and under I3 (the corroboration path) is unchanged; only the name of the monitored region changes.

7. Conformance

This instrument governs the term "envelope." Restatements shall be conformed at each instrument's next revision, and the tier rule at §2.1 applies whether or not an instrument is named here.

Conformance is staged, following the model by which the "Bright Line" reservation was rolled out per document rather than all at once:

1. **The governance core, immediately.** FD-BL §6.3 becomes a pointer; FD-BL-D1 and FD-BL-D2 cite FD-EV and drop their envelope holds. These already use the reserved sense, so this is citation and annotation, not rewording.
2. **The engineering annexes, per revision.** Instruments using "envelope" in the operational sense — chiefly the DBA-GAS series (especially FC2), the DBA-MA-SMR / nuclear set, and the electric-sector annexes — rename Sense O to "operating band" and carry the one-line reservation notice, each at its next revision. The execution plan enumerates the exact sites and counts before any edit; no rename is applied blind.
3. **New instruments** adopt the reservation from the declaration convention (§2.3) at authoring.

The detailed site list, replacement text, and sequencing are carried in the FD-EV amendment set (the execution plan and draft replacement text), not reproduced here.

8. Author Decisions Outstanding

One item closed in the October 2026 Edition. Confirmation of the renamed term against sector preference. The author confirmed "operating band" on 2 October 2026, and the permission for an established sector term, which this entry carried, is now a rule at §4.3.

Open: whether the operational-sense rename in the nuclear submittal set is coordinated with the DOI republication so a single re-issue carries both, rather than two revision cycles.

9. Foundational References

FD-BL — The Bright Line (placement, granularity, scope, obligation). FD-LD — layer decomposition and determinism. FD-BL-D1 — mode of discharge (uses the pre-authorized envelope). FD-BL-D2 — indication integrity. FD-LS — local sovereignty (§3.2 defines "governed act"). FD-BR — the Broker. The "Bright Line" reservation notices in the DBA-OIL and DBA-GAS series (the precedent this instrument follows).

Revision History

0.2 — 2 October 2026. Issued in the October 2026 Edition, and recorded in FD-REG v0.22 in the same act. One entry for the step of the Action Plan for the October 2026 Edition that opens this instrument (Step 7).

Step 7, 2 October 2026 (F-14, F-25, F-27; F-12 and F-24 in the past frame; Finding S6-1; F-22). **F-14:** §1 quotes FD-BL §6.3 as FD-BL v0.2 recorded it. **F-27, on the author's decision of 2 October 2026:** "operating band" is confirmed, the permission for an established sector term moves from §8 into §4.3 as a rule, and that §8 entry is closed; the second §8 item, on the nuclear submittal set, stays open. **F-25:** APC, SIS and MOC are expanded at first use, outside the quotation from DBA-GAS-FC2. **Finding S6-1:** §4.1 cites FD-LS §3.2 for "governed act," which this instrument used without defining. Under Decision C, §3 names the qualified human who defines an envelope as the Command Broker discharging in advance. §1 and §6 are put in the past frame where they described the state at first issue. §9 adds FD-LS and FD-BR. No change to the reservation, the rename or the test.

Verification read, 2 October 2026. §5 cites "the FD-BL §4.5 authorization" in full. No change to placement, granularity, scope or obligation.

Review of 2 October 2026. On the author's instruction to apply the cold review of the October 2026 Edition review draft as recommended (FD-BL Reconciliation/Independent_Review_2026-10-02_FD_October_2026_Edition_Review_Draft_v0_1), this instrument takes findings G-10, G-14, G-17; each edit is quoted in the review's application record.

0.1 — August 2026. Initial issue, resolving FD-BL v0.2 §6.3 and completing the FD-BL §6 set. Reserves "envelope" and "pre-authorized envelope" for the authorization sense (Sense G) used by FD-BL-D1 and FD-BL-D2; renames the operational sense (Sense O) to "operating band," with "operating limit," "encroachment," and "operating-limit excursion" for its boundary, approach, and violation; states the test distinguishing the two by the kind of obligation in view; issues as a Foundational Definition companion to FD-BL and FD-LD; and stages the conformance sweep (governance core immediately; engineering annexes per revision) following the "Bright Line" reservation model.